1 Sunflower detection using scale-space LoG

NAHG Nettikumara | 230436X | EN3160 — Fitting and Alignment

GitHub: https://github.com/hasith2003 · Assignment repository

The supplied JPEG was loaded with IMREAD_REDUCED_COLOR_4 (360 × 360 pixels). Dark flower centres were detected in grayscale I, scaled to [0,1]. The response $L(x,y,\sigma) = \sigma^2\nabla^2(G\sigma * I)$ was evaluated at 35 logarithmically spaced scales, σ = 1.5–18 pixels.

Positive local maxima were selected in a 3 × 3 × 3 position/scale neighbourhood above 0.1; boundary scales and circles crossing the image edge were rejected. A yellow-petal prior requires mean $\max(0,\min(R,G)-B) \geq 0.4$ in the annulus r to 1.9r. Greedy suppression rejects candidates within $0.8(r_1+r_2)$ of a stronger detection. Radius $r = \sqrt{2}\sigma$ gives 60 candidate central disks.

Scale-normalized LoG: 60 dark-centre detections



Rank	x (px)	y (px)	σ (px)	Radius (px)
1	282	338	6.96	9.84
2	106	255	6.01	8.51
3	179	327	6.01	8.51
4	178	261	5.59	7.91
5	189	240	4.49	6.35

```
response = sigma**2 * gaussian_laplace(gray, sigma)
peaks = (stack == maximum_filter(stack, size=(3,3,3))) & (stack > .10)
radius = np.sqrt(2) * sigma
```

The red labels identify the five largest retained circles in decreasing radius. Coordinates use the reduced image, with origin at the top left; original-image coordinates and radii are approximately four times larger. Circles describe the dark central disks rather than the outer petal boundary. Leaves surrounded by yellow petals, touching flowers and occlusion can produce false positives or missed centres. These are detections, not a validated flower count.

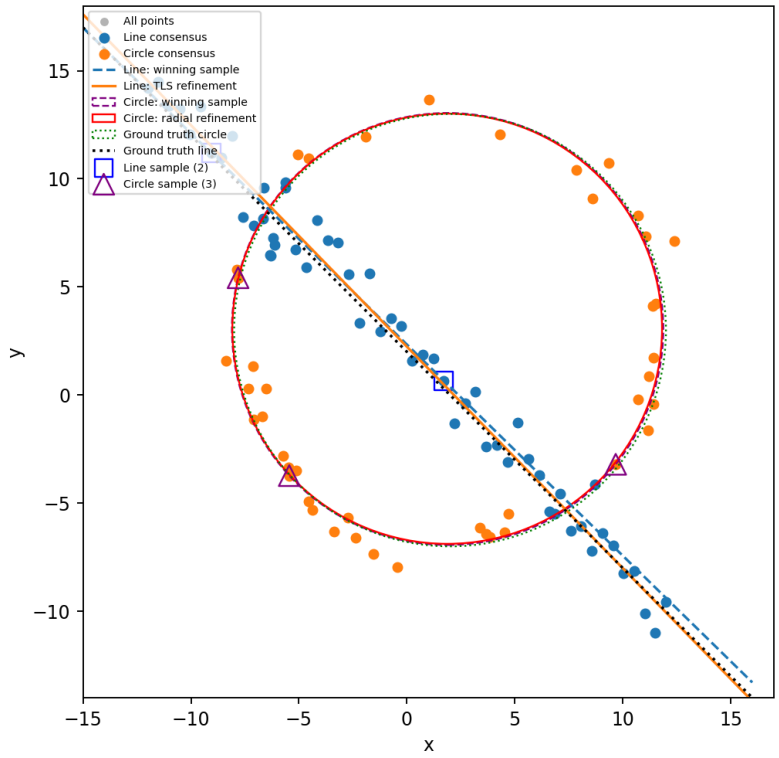
2 Sequential RANSAC: line, then circle

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Listing 1 was reproduced with seed 0: 50 noisy circle points and 50 noisy line points. Custom RANSAC samples two points for a line and three for a circle, rejecting degenerate samples. The line is ax + by = d with $a^2+b^2 = 1$ and $d \geq 0$. Line error is $|ax+by-d|$; circle error is $|\|p-c\|-r|$.

Both thresholds are 1.5 coordinate units; 5,000 hypotheses are tried per model, with minimum consensuses of 30 (line) and 25 (circle). Largest consensus wins; truncated squared error breaks ties. The winning line consensus is removed before circle fitting. Total least squares refines the line; bounded radial least squares refines the circle on its winning consensus.

Model	Refined parameters	Consensus
Line	a=0.7156, b=0.6985, d=1.5618	57
Circle	cx=1.8696, cy=3.0655, r=9.9493	43




```
line = ransac(X, 'line', seed=10)
remaining = X[~line[2]]
circle = ransac(remaining, 'circle', seed=20, min_consensus=25)
```

The line consensus includes 7 true circle points near the intersections. The reported consensuses are the winning sample's sets used for refinement; membership is not relabelled afterward. If the circle is fit first, this run returns centre (2.06, 2.84), radius 9.91, with 61 consensus points including 12 true line points.

A large-radius circle can approximate a noisy line over a short arc, and a flexible circle can also absorb points near line/circle intersections. Thus fitting the circle first can remove useful line points or select a spurious large circle. It need not fail every time: the outcome depends on noise, thresholds and samples.

3 Planar image compositing

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Four corners were selected manually from each displayed background and stored in top-left, top-right, bottom-right, bottom-left order. A homography H maps artwork corners to these points: $\lambda[x',y',1]^T = H[x,y,1]^T$. cv.getPerspectiveTransform solves the four correspondences; cv.warpPerspective resamples the artwork and a support mask into the photograph.

Both examples advertise imaginary space tourism: a Mars poster in a street display, and The Grand Tour on a wall display. The portrait Grand Tour artwork is centred on a cream landscape canvas before warping to preserve its proportions. Inside the warped mask, I = 0.94W + 0.06B; outside it, I = B. The supplied Wadham/flag example was also reproduced.

Selected corners



Source artwork



Mars street display



Selected corners



Source artwork



Grand Tour wall display




```
H = cv.getPerspectiveTransform(source_corners, target_corners)
W = cv.warpPerspective(artwork, H, background_size)
M = cv.warpPerspective(np.ones((h,w), np.float32), H, background_size)
result = background*(1 - .94*M[...,None]) + .94*W*M[...,None]
```

The displays are nearly frontal, so these examples show modest projective distortion. A single homography assumes one plane and does not model fabric wrinkles, surface curvature or foreground occluders. Alpha blending retains a little background illumination but does not physically reproduce shadows, glare or folds. Corner selection is a source of edge-placement error.

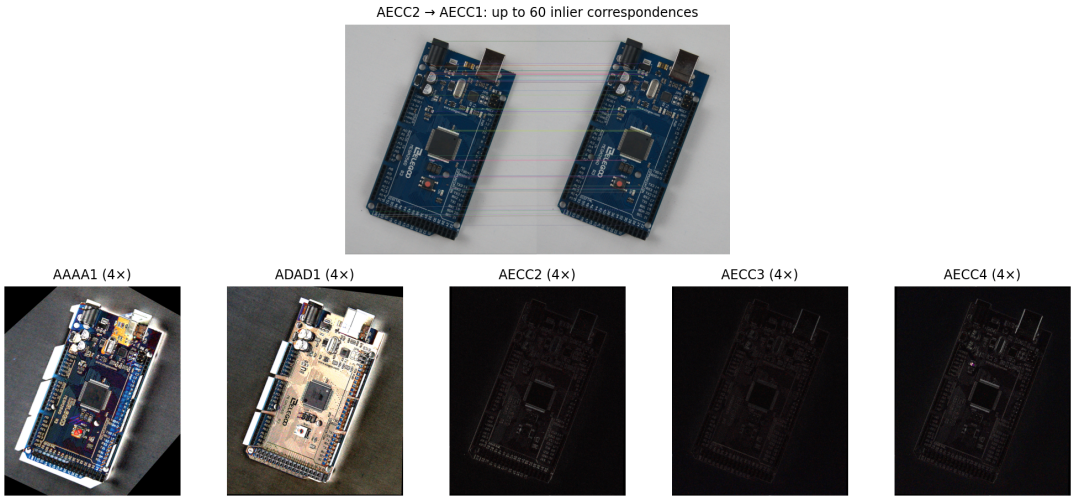
Backgrounds: Julia Filirovska, Pexels 4913828; Timothy Huliselan, Pexels 27467330. Artwork: NASA/JPL, Mars and The Grand Tour. Downloaded 19 October 2026; full URLs and credits: my_images/SOURCES.md.

4 PCB feature alignment and differences

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AECC1.jpg is the template. SIFT (up to 12,000 features) is applied to grayscale images at their original resolution. BFMatcher with L2 distance finds two neighbours; Lowe's ratio test retains $d_1 < 0.75 d_2$. A RANSAC homography maps each input to template coordinates using a 3-pixel reprojection threshold, up to 20,000 iterations and confidence 0.999.

The input is warped to the template size. Raw absolute BGR differences $D = |I_{aligned} - I_{template}|$ are computed in uint8. A separately warped support mask, eroded by 3 pixels, excludes unsupported borders from the figures and mean absolute error (MAE). MAE is averaged across valid pixels and colour channels; it is an intensity difference in [0,255], not a defect rate.



Top: AECC2 inlier matches. Bottom: all five non-template differences, displayed at 4× gain for visibility; raw lossless maps are saved separately.

Image	Ratio matches	H inliers	Median error px	Overlap MAE
AAAA1.jpg	57	14	1.376	28.99
ADAD1.jpg	339	190	0.775	30.63
AECC2.jpg	4095	3583	0.366	3.52
AECC3.jpg	5071	4509	0.304	3.30
AECC4.jpg	3608	3167	0.403	4.07

```
good = [a for a,b in matcher.knnMatch(desc, template_desc, k=2)
        if a.distance < .75*b.distance]
H, inliers = cv.findHomography(src, dst, cv.RANSAC, 3.0)
difference = cv.absdiff(cv.warpPerspective(image, H, size), template)
```

Large differences on the USB Type B port, capacitors and reset button can arise from height above the PCB plane: viewpoint changes cause parallax that one planar homography cannot remove. Specular reflections on metal, shadows, illumination changes and occlusion also contribute. Real component/layout differences are another possibility; the difference map alone cannot separate these causes.

AAAA1 has fewer stable correspondences and a much smaller inlier fraction than the AECC images, so its alignment deserves extra caution. Low inlier reprojection error supports alignment of matched features, but does not prove every component is registered. A global image MAE also includes background and lighting changes. Reliable defect detection would need board-region masking, photometric normalization and local/component-level verification.